

Clinical Pathway Handbook: Differentials and Management by Condition

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Format note: Management text is kept close to source wording and reorganized into pathway branches for RAG use.

Safety note: This is decision support content; clinician judgment and local policy always take priority.

1. Condition 2: Pregnant Patient Under 20 Weeks Gestation When a pregnant patient presents with per vaginal bleeding before 20 weeks' gestation, manage as threatened, incomplete, inevitable, missed, septic, or complete abortion until ectopic pregnancy has been excluded by obstetric ultrasound and serum human chorionic gonadotropin (hCG) trend

Clinical framing:

Condition 2: Pregnant Patient Under 20 Weeks Gestation When a pregnant patient presents with per vaginal bleeding before 20 weeks' gestation, manage as threatened, incomplete, inevitable, missed, septic, or complete abortion until ectopic pregnancy has been excluded by obstetric ultrasound and serum human chorionic gonadotropin (hCG) trend. ---

Key differentials to consider:

- Condition-specific differentials were not explicitly enumerated in the source text; use local protocol-based differential diagnosis.

Pathway branches and management:

- If the patient is shocked, initiate immediate airway, breathing, and circulation support using the ABCDE approach; secure intravenous access; administer 1 litre of 0.9% sodium chloride over 15 minutes; send urgent complete blood count, blood group and cross-match, renal function and electrolytes, and arrange urgent obstetric ultrasound.
- When obstetric ultrasound confirms retained products of conception and the patient is haemodynamically stable, medical or surgical evacuation should be performed under gynaecological supervision.
- When granulocytosis is present alongside fever, uterine tenderness, or offensive vaginal discharge, suspect septic abortion and administer intravenous ceftriaxone and metronidazole while arranging evacuation.
- When thrombocytopenia or coagulopathy is identified, urgently assess for disseminated intravascular coagulation and involve haematology and obstetric specialists.

Follow-up question guidance:

If branch selection is uncertain, ask one focused question on severity, hemodynamic status, red flags, pregnancy status, age group, or contraindications before final management step.

2. Condition 3: Pregnant Patient At or Beyond 20 Weeks Gestation When a pregnant patient at or beyond 20 weeks' gestation presents with painless per vaginal bleeding, placenta previa must be suspected until excluded by transabdominal and transvaginal obstetric ultrasound

Clinical framing:

Condition 3: Pregnant Patient At or Beyond 20 Weeks Gestation When a pregnant patient at or beyond 20 weeks' gestation presents with painless per vaginal bleeding, placenta previa must be suspected until excluded by transabdominal and transvaginal obstetric ultrasound. Immediate resuscitation, urgent obstetric ultrasound, continuous fetal monitoring, and preparation for emergency caesarean delivery are required—even if ultrasound appears normal, as up to 10% of abruptions are sonographically occult.

Key differentials to consider:

- peptic ulcer disease
- pancreatitis
- biliary colic/cholecystitis
- myocardial ischemia
- gastritis/GERD

Pathway branches and management:

- If placenta previa is confirmed and bleeding is active or recurrent, admit immediately, restrict vaginal examination, avoid rectal interventions, monitor fetal heart rate continuously, administer intravenous tranexamic acid within 3 hours of presentation, and prepare for emergency delivery if maternal or fetal compromise develops.
- When bleeding is associated with uterine tenderness, abnormal fetal heart rate, or rising maternal pulse and falling blood pressure, placental abruption must be suspected.
- If the patient develops eclamptic features—including severe headache, visual disturbance, epigastric pain, altered mental status, or seizure—treat as eclampsia: protect airway, administer intravenous magnesium sulfate per local protocol, control severe hypertension, and obtain urgent obstetric review.

Follow-up question guidance:

If branch selection is uncertain, ask one focused question on severity, hemodynamic status, red flags, pregnancy status, age group, or contraindications before final management step.

3. Condition 4: Postpartum Haemorrhage When a patient bleeds more than 500 millilitres of blood within 24 hours of delivery—or more than 1000 millilitres after caesarean delivery—postpartum haemorrhage is diagnosed**Clinical framing:**

Condition 4: Postpartum Haemorrhage When a patient bleeds more than 500 millilitres of blood within 24 hours of delivery—or more than 1000 millilitres after caesarean delivery—postpartum haemorrhage is diagnosed. Condition 5: Sepsis-Associated Per Vaginal Bleeding When per vaginal bleeding occurs alongside fever exceeding 38.0 degrees Celsius, tachycardia above 90 beats per minute, tachypnoea above 22 breaths per minute, or altered mental status, sepsis must be suspected. Condition 6: Coagulopathy-Related Bleeding When per vaginal bleeding is accompanied by thrombocytopenia (platelet count below 150×10^9 per litre), prolonged activated partial thromboplastin time, or low fibrinogen (<2 grams per litre), coagulopathy must be investigated.

Key differentials to consider:

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Pathway branches and management:

- If uterine atony is suspected (a soft, boggy uterus), perform bimanual uterine massage and administer intramuscular oxytocin 10 international units.
- If bleeding persists, give intravenous ergometrine 0.2 milligrams (unless contraindicated by hypertension) or intravenous carboprost 250 micrograms.
- When genital tract trauma is identified, repair lacerations under adequate analgesia or anaesthesia.
- When retained placental tissue is suspected, perform manual removal of placenta under appropriate analgesia and antibiotic prophylaxis.
- If shock develops—defined as systolic blood pressure below 90 millimetres of mercury, tachycardia exceeding 120 beats per minute, altered mental status, or oliguria—administer 2 litres of intravenous crystalloid over 30 minutes, transfuse packed red blood cells based on haemoglobin and ongoing loss,

and activate massive transfusion protocol if indicated..

- If the patient has uterine tenderness, offensive vaginal discharge, or history of recent instrumentation or abortion, intravenous broad-spectrum antibiotics—ceftriaxone 2 grams and metronidazole 500 milligrams—are indicated immediately, without waiting for culture results.
- When granulocytosis is present with thrombocytopenia or acute kidney injury, consider septic shock and initiate early goal-directed therapy: central venous pressure monitoring, vasopressor support if mean arterial pressure remains below 65 millimetres of mercury after 3 litres of fluid, and source control—including evacuation of infected retained products..
- If disseminated intravascular coagulation is suspected—characterised by concurrent thrombocytopenia, elevated D-dimer, prolonged prothrombin time, and low fibrinogen—administer fresh frozen plasma, cryoprecipitate, and platelet transfusion according to local massive transfusion protocol and haematology guidance.
- When von Willebrand disease is suspected in a young woman with lifelong menorrhagia and mucocutaneous bleeding, measure von Willebrand factor antigen, ristocetin cofactor activity, and factor VIII level—and refer to haematology before initiating desmopressin or von Willebrand factor concentrate..
- If systolic blood pressure is 160 millimetres of mercury or higher, or diastolic blood pressure is 110 millimetres of mercury or higher, administer intravenous labetalol or hydralazine to reduce mean arterial pressure gradually—avoiding precipitous drops that may compromise placental perfusion.
- When severe features are present—including impaired consciousness, pulmonary oedema, oliguria, thrombocytopenia, elevated transaminases, or renal impairment—urgent delivery is indicated regardless of gestational age.

Follow-up question guidance:

If branch selection is uncertain, ask one focused question on severity, hemodynamic status, red flags, pregnancy status, age group, or contraindications before final management step.

4. Condition 8: Head Injury When a patient presents with head injury and Glasgow Coma Scale score below 8, secure the airway with endotracheal intubation, prevent hypoxia and hypotension, administer intravenous mannitol or hypertonic saline only if signs of cerebral herniation are present, and arrange urgent non

Clinical framing:

Condition 8: Head Injury When a patient presents with head injury and Glasgow Coma Scale score below 8, secure the airway with endotracheal intubation, prevent hypoxia and hypotension, administer intravenous mannitol or hypertonic saline only if signs of cerebral herniation are present, and arrange urgent non-contrast computed tomography of the head. ---

Key differentials to consider:

- intracranial hemorrhage
- concussion
- skull fracture
- cervical spine injury
- raised intracranial pressure

Pathway branches and management:

- If Glasgow Coma Scale score is between 8 and 12, admit for observation, repeat neurological assessments every 30 minutes initially, and obtain urgent neuroimaging and neurosurgical consultation.
- When Glasgow Coma Scale score is 13 or higher but red flags are present—including vomiting, convulsions, confusion, cerebrospinal fluid leak, anisocoria, or lateralising neurological signs—admit for 24-hour observation and serial assessments.

Follow-up question guidance:

If branch selection is uncertain, ask one focused question on severity, hemodynamic status, red flags, pregnancy status, age group, or contraindications before final management step.

5. Condition 9: Abdominal Injury When a patient presents with blunt abdominal trauma and signs of shock—tachycardia above 120 beats per minute, systolic blood pressure below 90 millimetres of mercury, altered mental status, or poor capillary refill—initiate immediate intravenous resuscitation with two large

Clinical framing:

Condition 9: Abdominal Injury When a patient presents with blunt abdominal trauma and signs of shock—tachycardia above 120 beats per minute, systolic blood pressure below 90 millimetres of mercury, altered mental status, or poor capillary refill—initiate immediate intravenous resuscitation with two large-bore cannulae and 2 litres of 0.9% sodium chloride, perform focused assessment with sonography in trauma (FAST) exam, and prepare for emergency laparotomy or transfer. ---

Key differentials to consider:

- solid organ injury
- hollow viscus injury
- intra-abdominal hemorrhage
- peritonitis
- retroperitoneal injury

Pathway branches and management:

- If penetrating abdominal trauma is present with an impaled object, do not remove it—stabilise the object with dressings and immobilise the patient during transport.
- If no impaled object but external bleeding is present, apply direct pressure with sterile gauze and administer intravenous broad-spectrum antibiotics per local trauma protocol.
- When abdominal pain persists or examination reveals rigidity, rebound tenderness, or absent bowel sounds in a stable patient, admit for serial abdominal examinations, repeat FAST, and surgical review within 6 hours.

Follow-up question guidance:

If branch selection is uncertain, ask one focused question on severity, hemodynamic status, red flags, pregnancy status, age group, or contraindications before final management step.

6. Condition 10: Wheeze and Acute Asthma Exacerbation When a patient presents with wheeze and respiratory distress, stratify severity by ability to speak, respiratory rate, accessory muscle use, oxygen saturation, and mental status

Clinical framing:

Condition 10: Wheeze and Acute Asthma Exacerbation When a patient presents with wheeze and respiratory distress, stratify severity by ability to speak, respiratory rate, accessory muscle use, oxygen saturation, and mental status. ---

Key differentials to consider:

- acute asthma exacerbation
- COPD/reactive airway disease
- anaphylaxis
- pneumonia with bronchospasm
- cardiac asthma

Pathway branches and management:

- If the patient is in severe exacerbation—unable to complete sentences, respiratory rate above 30 breaths per minute, oxygen saturation below 92% on room air, or pulsus paradoxus above 25 millimetres of mercury—administer high-dose inhaled salbutamol via oxygen-driven nebuliser, intravenous magnesium sulfate 1.2–2 grams over 20 minutes, and oral or intravenous corticosteroids.
- If impending respiratory failure is suspected—fatigue, drowsiness, confusion, bradycardia, or silent chest—prepare for assisted ventilation and urgent intensive care referral.

Follow-up question guidance:

If branch selection is uncertain, ask one focused question on severity, hemodynamic status, red flags, pregnancy status, age group, or contraindications before final management step.

7. Condition 11: Fever in Children and Adults When a child older than 60 days presents with fever lasting less than 7 days and no danger signs, assess hydration, malaria risk, respiratory symptoms, urinary symptoms, rash, neck stiffness, and immunosuppression

Clinical framing:

Condition 11: Fever in Children and Adults When a child older than 60 days presents with fever lasting less than 7 days and no danger signs, assess hydration, malaria risk, respiratory symptoms, urinary symptoms, rash, neck stiffness, and immunosuppression. ---

Key differentials to consider:

- Condition-specific differentials were not explicitly enumerated in the source text; use local protocol-based differential diagnosis.

Pathway branches and management:

- If complete blood count is normal and malaria antigen test is negative, consider viral or atypical infection and offer outpatient management with antipyretics, oral rehydration, and clear safety-netting instructions—including return if fever persists beyond 5 days, develops rash, or shows signs of dehydration or lethargy.
- If malaria test is positive and the patient is stable with oral intake, treat as outpatient with weight-based artemisinin-based combination therapy per national policy.
- If severe malaria is suspected—impaired consciousness, repeated seizures, shock, severe anaemia, or acidosis—admit immediately, start parenteral artesunate, monitor glucose and renal function, and request urgent senior review.

Follow-up question guidance:

If branch selection is uncertain, ask one focused question on severity, hemodynamic status, red flags, pregnancy status, age group, or contraindications before final management step.

8. Condition 12: Hyperkalaemia When serum potassium exceeds 5

Clinical framing:

Condition 12: Hyperkalaemia When serum potassium exceeds 5.5 millimoles per litre and electrocardiographic changes are present—including peaked T waves, flattened P waves, widened QRS complex, or sine wave pattern—administer intravenous calcium gluconate 10% (10 millilitres over 2–5 minutes) to stabilise the myocardium. ---

Key differentials to consider:

- Condition-specific differentials were not explicitly enumerated in the source text; use local protocol-based differential diagnosis.

Pathway branches and management:

- If serum potassium exceeds 6.0 millimoles per litre without ECG changes, shift potassium into cells using intravenous insulin 10 units with 50 millilitres of 50% dextrose, nebulised salbutamol 10 puffs, and intravenous sodium bicarbonate 50 millilitres of 8.4% if metabolic acidosis is confirmed.
- When potassium remains elevated despite these measures—or if renal failure is present—initiate potassium removal with oral potassium binders (e.g., sodium zirconium cyclosilicate), loop diuretics if volume overloaded, or urgent haemodialysis.

Follow-up question guidance:

If branch selection is uncertain, ask one focused question on severity, hemodynamic status, red flags, pregnancy status, age group, or contraindications before final management step.

9. Condition 13: Diabetic Ketoacidosis When a patient presents with polyuria, polydipsia, nausea, vomiting, abdominal pain, Kussmaul breathing, fruity breath odour, blood glucose above 11 millimoles per litre, ketonaemia or ketonuria, and arterial pH below 7

Clinical framing:

Condition 13: Diabetic Ketoacidosis When a patient presents with polyuria, polydipsia, nausea, vomiting, abdominal pain, Kussmaul breathing, fruity breath odour, blood glucose above 11 millimoles per litre, ketonaemia or ketonuria, and arterial pH below 7.3 or serum bicarbonate below 15 millimoles per litre—diabetic ketoacidosis is diagnosed. After initial resuscitation, begin intravenous insulin infusion at 0.1 units per kilogram per hour, continue isotonic saline until corrected sodium normalises, and switch to 0.45% saline when serum sodium falls below 135 millimoles per litre to avoid cerebral oedema. Monitor potassium closely—supplement intravenously if serum potassium falls below 3.3 millimoles per litre, even if total body potassium is high.

Key differentials to consider:

- Condition-specific differentials were not explicitly enumerated in the source text; use local protocol-based differential diagnosis.

Pathway branches and management:

- If shock is present—systolic blood pressure below 90 millimetres of mercury, tachycardia above 120 beats per minute, or altered mental status—administer 1–2 litres of 0.9% sodium chloride over 30–60 minutes before starting insulin infusion.

Follow-up question guidance:

If branch selection is uncertain, ask one focused question on severity, hemodynamic status, red flags, pregnancy status, age group, or contraindications before final management step.

10. Condition 14: Burns When a patient presents with partial

Clinical framing:

Condition 14: Burns When a patient presents with partial-thickness or full-thickness burns involving more than 10% total body surface area in adults—or more than 5% in children—immediate transfer to a burn centre is indicated. Administer high-flow oxygen, establish intravenous access proximal to burned areas, and calculate fluid resuscitation using the modified Parkland formula: 2–4 millilitres of 0.9% sodium chloride per kilogram per percentage total body surface area burned over 24 hours—with half given in the first 8 hours. Do not apply ice, butter, or home remedies.

Key differentials to consider:

- inhalation injury
- electrical burn complications

- sepsis in burn wounds
- compartment syndrome
- dehydration/hypovolemia

Pathway branches and management:

- If facial burns are present, assess airway patency urgently—consider early intubation if stridor, hoarseness, or soot in sputum is noted.

Follow-up question guidance:

If branch selection is uncertain, ask one focused question on severity, hemodynamic status, red flags, pregnancy status, age group, or contraindications before final management step.

11. Condition 15: Upper Gastrointestinal Bleeding When a patient presents with haematemesis, melaena, or haematochezia and signs of shock—systolic blood pressure below 90 millimetres of mercury, tachycardia above 120 beats per minute, or altered mental status—secure two large

Clinical framing:

Condition 15: Upper Gastrointestinal Bleeding When a patient presents with haematemesis, melaena, or haematochezia and signs of shock—systolic blood pressure below 90 millimetres of mercury, tachycardia above 120 beats per minute, or altered mental status—secure two large-bore intravenous lines, administer 1–2 litres of 0.9% sodium chloride rapidly, cross-match 4 units of packed red blood cells, and prepare for urgent upper gastrointestinal endoscopy.

Key differentials to consider:

- Condition-specific differentials were not explicitly enumerated in the source text; use local protocol-based differential diagnosis.

Pathway branches and management:

- If haemoglobin is below 10 grams per decilitre and bleeding is active, transfuse packed red blood cells to maintain haemoglobin above 7–8 grams per decilitre in most patients—or above 9–10 grams per decilitre in those with cardiovascular disease.
- When tranexamic acid is used, administer intravenous loading dose of 1 gram over 10 minutes, followed by 1 gram infusion over 8 hours—only if endoscopy can be performed within 8 hours and bleeding is not due to varices or malignancy.
- If *Helicobacter pylori* testing is positive, initiate eradication therapy with proton pump inhibitor, amoxicillin, and clarithromycin per local antimicrobial guidelines—not ranitidine.

Follow-up question guidance:

If branch selection is uncertain, ask one focused question on severity, hemodynamic status, red flags, pregnancy status, age group, or contraindications before final management step.

RAG Response Contract (recommended)

Return concise output with: identified condition, selected branch, next safest management step, and one clarifying question only when required.

Schema: assistant_reply, next_actions, clarification_questions, insufficient_evidence.